



Machine Learning

Quiz - 1 (20 Marks) (Duration 60 min*)



Beginners (8 marks)

- The process of minimizing the average error calculated on the training dataset is formally known as...
A. Maximum Likelihood Estimation B. Empirical Risk Minimization C. Loss optimization D. Regularization
B
- Probit regression is an alternative to logistic regression that assumes an underlying...
A. Poisson B. Gamma C. Normal D. Binomial
C
- You flip a coin 10 times and get 7 heads. What is the MLE for the probability of getting heads?
A. 3 B. 1.4 C. 7 D. 0.7
D
- If logistic regression outputs probabilities bounded between 0 and 1, why can't we simply use linear regression and clip the outputs to the range?
Clipping creates discontinuous gradients at the boundaries (0 and 1), making optimization difficult. Linear regression assumes Gaussian noise, which is inappropriate for binary outcomes. The sigmoid function provides smooth, differentiable transitions and has theoretical justification through MLE for Bernoulli distribution.
- If $h_b(x_i) = 1$ when $x_i b > 0$, what class would be predicted when $x_i b = 0$ exactly?
When $x_i b = 0$ exactly, the point lies on the decision boundary. The hypothesis function would predict Class 0 (since it's "0 otherwise"), but this is an edge case that rarely occurs in practice with real-valued data. The sigmoid probability would be exactly 0.5, indicating maximum uncertainty.
- How can we reduce/avoid overfitting from a function approximation perspective?
Reducing the space; Making the choice of f^* less dependent on data
- As the amount of training data becomes very large, the difference between the MAP and MLE estimates typically... (and Why?)
A. Increases B. Stays the same C. Decreases D. Can't be determined
C; With enough data, the likelihood term overwhelms the prior, and the MAP estimate converges to the MLE estimate.
- Explain Berkson's paradox.
Check GC

Intermediate (2 marks)

- A programmer writes a set of explicit, hard-coded rules to determine if an email is spam (e.g., "IF the email contains the words 'free money', THEN mark as spam"). A machine learning engineer, however, feeds a model 100,000 emails that are already labeled as "spam" or "not spam" and lets the model learn the patterns itself. Why is the machine learning engineer's approach considered inductive learning, while the programmer's approach is not?
The ML engineer's approach is inductive because the model is deriving a general rule (what makes an email spam) from specific examples (the labelled emails). It induces the pattern from the data. The programmer's approach is based on pre-defined rules and does not learning from data; it simply executes commands.
- Model A:** is a very simple linear model that predicts a student's final grade based only on the number of classes they attended. It performs poorly on both the training and test datasets. **Model B** is a highly complex model that uses 100 different features (like the student's shoe size, breakfast habits, etc.). It has 100% accuracy on the training data but performs very poorly on the test data. Comment on the bias/variance of the models and link your reasoning to underfitting and overfitting.
Model A suffers from high bias and is underfitting. Model B suffers from high variance and is overfitting.

3. A model is designed to detect fraudulent bank transactions. In a test with 1,000 transactions, there were 10 actual fraudulent transactions. The model produced the following results: Correctly identified 8 of the fraudulent transactions; Incorrectly flagged 5 legitimate transactions as fraud; Missed 2 of the fraudulent transactions. In this scenario, which is metric (such as acc, precision, TNR, F1-score etc...) is more important for the bank?

A bank would rather annoy a few customers with false alarms (low precision) than miss fraudulent transactions (low recall), which could cost millions. The goal is to catch as much fraud as possible, making recall the critical metric.

4. A city analyst is tasked with predicting the number of bicycles rented from a station each hour. They start with a standard linear regression model but notice a problem: for certain times of day with low traffic, the model predicts -3 bicycles will be rented, which is physically impossible. Why is standard linear regression failing here? How do we fix this? (Hint: Think probabilistically!)

Standard linear regression fails because it assumes the output can be any real number and that its errors follow a normal (Gaussian) distribution. Count data (like the number of bikes) cannot be negative and often follows a different distribution, like a Poisson distribution, hence we require a GLM.

Advanced (4 marks)

1. Derive MLE for logistic regression when the assumption is made that the target variable originates from a Gaussian distribution.

The probability density function of a Gaussian distribution is:

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Now if we consider $y = 1$, the probability becomes:

$$P(y = 1|x) = \frac{P(x|y = 1)P(y = 1)}{P(x|y = 1)P(y = 1) + P(x|y = 0)P(y = 0)} \quad (1)$$

For $y = 1$, we have Gaussian with mean μ_a and variance σ_a^2 , and for $y = 0$, we have Gaussian with mean μ_b and variance σ_b^2 . Thus:

$$P(x|y = 1) = \frac{1}{\sqrt{2\pi\sigma_a^2}} e^{-\frac{(x-\mu_a)^2}{2\sigma_a^2}} \quad (2)$$

$$P(x|y = 0) = \frac{1}{\sqrt{2\pi\sigma_b^2}} e^{-\frac{(x-\mu_b)^2}{2\sigma_b^2}} \quad (3)$$

Let $P(y = 1) = p$ and $P(y = 0) = 1 - p$. Now, putting these into the equation above:

$$P(y = 1|x) = \frac{\frac{1}{\sqrt{2\pi\sigma_a^2}} e^{-\frac{(x-\mu_a)^2}{2\sigma_a^2}} p}{\frac{1}{\sqrt{2\pi\sigma_a^2}} e^{-\frac{(x-\mu_a)^2}{2\sigma_a^2}} p + \frac{1}{\sqrt{2\pi\sigma_b^2}} e^{-\frac{(x-\mu_b)^2}{2\sigma_b^2}} (1-p)} \quad (4)$$

For simplicity, let's assume $\sigma_a = \sigma_b = \sigma$, so the expression simplifies to:

$$P(y = 1|x) = \frac{1}{1 + e^{-\frac{(x-\mu_b)^2 - (x-\mu_a)^2}{2\sigma^2} \frac{1-p}{p}}} \quad (5)$$

$$= \frac{1}{1 + e^{-\frac{(x-\mu_b)^2 - (x-\mu_a)^2}{2\sigma^2}} \cdot e^{\log(\frac{1-p}{p})}} \quad (6)$$

$$= \frac{1}{1 + e^{\frac{-(x-\mu_b)^2 + (x-\mu_a)^2}{2\sigma^2} + \log(\frac{1-p}{p})}} \quad (7)$$

$$= \frac{1}{1 + e^{-\beta x + y}} \quad (8)$$

Observe the sigmoid-like function again, as you saw in derivation with the Bernoulli distribution.

2. Derivation of the Closed-Form Solution for Ridge Regression.

The objective function for Ridge Regression is the sum of the Residual Sum of Squares (RSS) and an L2 regularization penalty. In matrix notation, this is:

$$J(\mathbf{w}) = \|\mathbf{X}\mathbf{w} - \mathbf{y}\|_2^2 + \lambda \|\mathbf{w}\|_2^2$$

The squared L2 norm of a vector $\mathbf{v}$ is $\mathbf{v}^T \mathbf{v}$. Applying this to our objective function:

$$\begin{aligned} J(\mathbf{w}) &= (\mathbf{X}\mathbf{w} - \mathbf{y})^T (\mathbf{X}\mathbf{w} - \mathbf{y}) + \lambda \mathbf{w}^T \mathbf{w} \\ &= (\mathbf{w}^T \mathbf{X}^T - \mathbf{y}^T) (\mathbf{X}\mathbf{w} - \mathbf{y}) + \lambda \mathbf{w}^T \mathbf{w} \\ &= \mathbf{w}^T \mathbf{X}^T \mathbf{X} \mathbf{w} - \mathbf{w}^T \mathbf{X}^T \mathbf{y} - \mathbf{y}^T \mathbf{X} \mathbf{w} + \mathbf{y}^T \mathbf{y} + \lambda \mathbf{w}^T \mathbf{w} \end{aligned}$$

Since $\mathbf{y}^T \mathbf{X} \mathbf{w}$ is a scalar, it is equal to its own transpose, $(\mathbf{y}^T \mathbf{X} \mathbf{w})^T = \mathbf{w}^T \mathbf{X}^T \mathbf{y}$. We can combine the two cross-terms:

$$J(\mathbf{w}) = \mathbf{w}^T \mathbf{X}^T \mathbf{X} \mathbf{w} - 2\mathbf{w}^T \mathbf{X}^T \mathbf{y} + \mathbf{y}^T \mathbf{y} + \lambda \mathbf{w}^T \mathbf{w}$$

We differentiate $J(\mathbf{w})$ term by term using the following standard matrix calculus identities:

- $\nabla_{\mathbf{w}}(\mathbf{w}^T \mathbf{A} \mathbf{w}) = 2\mathbf{A} \mathbf{w}$ (for symmetric $\mathbf{A}$)
- $\nabla_{\mathbf{w}}(\mathbf{b}^T \mathbf{w}) = \mathbf{b}$

Note that $\mathbf{X}^T \mathbf{X}$ is a symmetric matrix.

$$\begin{aligned} \nabla_{\mathbf{w}} J(\mathbf{w}) &= \nabla_{\mathbf{w}} (\mathbf{w}^T \mathbf{X}^T \mathbf{X} \mathbf{w} - 2\mathbf{w}^T \mathbf{X}^T \mathbf{y} + \mathbf{y}^T \mathbf{y} + \lambda \mathbf{w}^T \mathbf{w}) \\ &= 2\mathbf{X}^T \mathbf{X} \mathbf{w} - 2\mathbf{X}^T \mathbf{y} + \mathbf{0} + 2\lambda \mathbf{I} \mathbf{w} \end{aligned}$$

Note: $\mathbf{w}^T \mathbf{w} = \mathbf{w}^T \mathbf{I} \mathbf{w}$, where $\mathbf{I}$ is the identity matrix. To find the minimum of the convex function $J(\mathbf{w})$, we set its gradient to the zero vector and solve for $\mathbf{w}$:

$$\begin{aligned} 2\mathbf{X}^T \mathbf{X} \mathbf{w} - 2\mathbf{X}^T \mathbf{y} + 2\lambda \mathbf{I} \mathbf{w} &= \mathbf{0} \\ \mathbf{X}^T \mathbf{X} \mathbf{w} - \mathbf{X}^T \mathbf{y} + \lambda \mathbf{I} \mathbf{w} &= \mathbf{0} \end{aligned}$$

We group the terms containing $\mathbf{w}$:

$$\begin{aligned} \mathbf{X}^T \mathbf{X} \mathbf{w} + \lambda \mathbf{I} \mathbf{w} &= \mathbf{X}^T \mathbf{y} \\ (\mathbf{X}^T \mathbf{X} + \lambda \mathbf{I}) \mathbf{w} &= \mathbf{X}^T \mathbf{y} \\ \mathbf{w}_{\text{ridge}} &= (\mathbf{X}^T \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^T \mathbf{y} \end{aligned}$$

This is the closed-form solution for the Ridge Regression coefficients. The term $\lambda \mathbf{I}$ ensures that the matrix is always invertible.

Bonus (2 marks)

1. A consistent estimator has the property that as the number of data points (which are used to estimate the parameters) increases, i.e. $N \rightarrow \infty$, the estimates converge in probability to the true parameters. Show that an unbiased estimator is not always consistent. Further, show that the OLS estimator is always consistent. You may use an example/counterexample to show the result.

We know an estimator is Unbiased when: $\mathbb{E}(\hat{\theta}_n) = \theta$, where $\hat{\theta}_n$ is the estimator for parameter θ . An estimator is said to be Consistent when:

$$\hat{\theta}_n \xrightarrow{p} \theta \quad \text{as } n \rightarrow \infty$$

i.e., the estimator converges to the true parameter as the number of observations increases. Example for i.i.d. data $X_1, X_2, \dots, X_n$ each having mean to be θ and variance to be σ . Let's say our estimator be a sample from above data points i.e. X_1 :

$$\hat{\theta}_n = X_1$$

which is unbiased, since: $\mathbb{E}[\hat{\theta}_n] = \mathbb{E}[X_1] = \theta$. Now we will check consistency. As $n \rightarrow \infty$, our variance of $\hat{\theta}_n$ does not go to zero since there is no effect of N on the variance for this estimator the variance remains same. The OLS (Ordinary Least Squares) estimator is a consistent estimator. We know that the OLS estimator for β is: $\hat{\beta} = (X^T X)^{-1} X^T Y$ where $Y = X\beta + \epsilon$ (error term), and $\mathbb{E}[\epsilon] = 0$. To prove that OLS is unbiased we will find the expectation of $\hat{\beta}$

$$\begin{aligned} \mathbb{E}[\hat{\beta}] &= \mathbb{E}[(X^T X)^{-1} X^T Y] \\ &= (X^T X)^{-1} X^T \mathbb{E}[Y] \\ &= (X^T X)^{-1} X^T \mathbb{E}[X\beta + \epsilon] \\ &= (X^T X)^{-1} X^T X \mathbb{E}[\beta] + (X^T X)^{-1} X^T \mathbb{E}[\epsilon] \\ &= \beta + 0 \\ \mathbb{E}[\hat{\beta}] &= \beta \end{aligned}$$

Thus, OLS is an unbiased estimator. Now to find if OLS is consistent, we will find the variance of the Sample Mean For data points $X_1, X_2, X_3, \dots, X_m$. having variance of each sample to be σ^2 . Variance of the average $\bar{X}_n$ is:

$$\begin{aligned}\text{Var}(\bar{X}_n) &= \frac{1}{N^2} \text{Var}(X_1 + X_2 + \dots + X_n) \\ &= \frac{1}{N^2} N\sigma^2 = \frac{\sigma^2}{N}\end{aligned}$$

As $N \rightarrow \infty$, $\text{Var}(\bar{X}_n) \rightarrow 0$, meaning $\hat{\beta}_n$ becomes more concentrated around β , showing consistency.

2. +1 for scoring ≥ 16 marks in the exam.